

Clay'sCars.com

“The all-in-one car finder”

System Requirements Documentation

Clay Vaughn-Creator

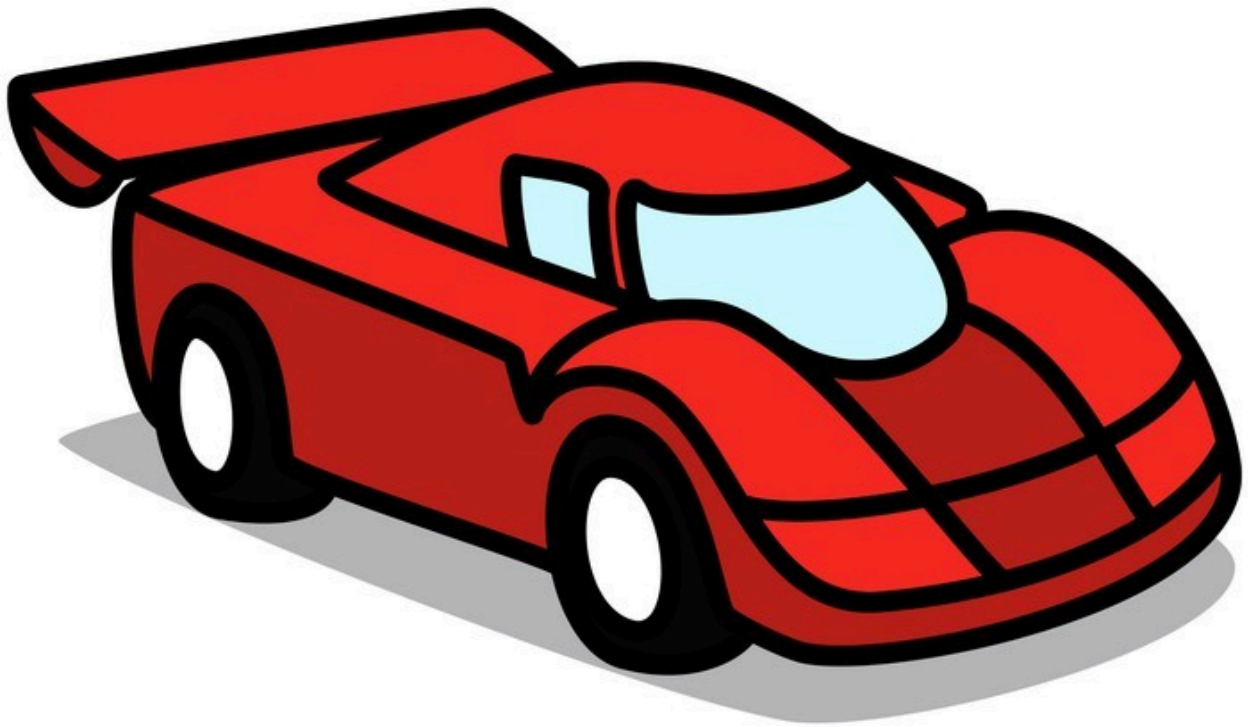


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Customer Problem Statements and System Requirements Clay'sCars.com

In all towns big and small we have car dealerships. Often selling different makes and models by manufacturer. These different individual dealerships may be many and often tedious to peruse through to find your next car. Each of these dealerships probably has its websites where you digitally peruse their inventory but still, it can be cumbersome to look at the websites of the 10 different dealerships in town.

Glossary of Terms-

- Latency- I wish to help slower internet connections by eliminating the need to load numerous websites.
- Quality Assurance- This system is a simple and easy way to look for cars locally.

Functional Requirements-

REQ-1, High, The System will source local dealership websites and display the specified cars the user inputted.

REQ-2, High, The System will allow specifications to be inputted to narrow down the retrieved data.

REQ-3, Medium, The System will have a nice display for user experience boosting.

REQ-4, High, The System will utilize a web API to be able to source local websites.

REQ-5, Low, The System will have a user login feature to display search history.

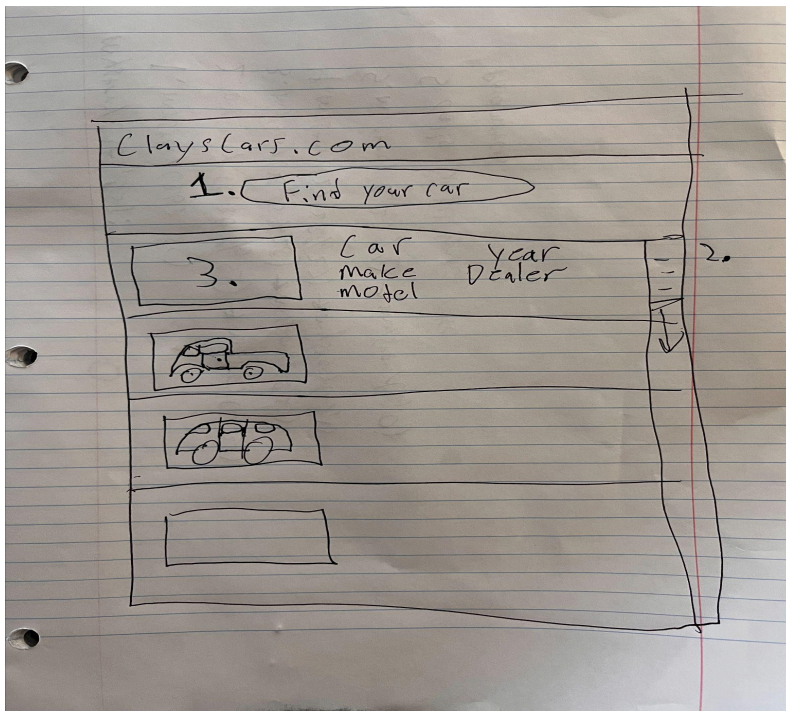
Nonfunctional Requirements-

1. Functionality-The size of the database will be dependent on the query of the user.
2. Usability- The user will have the ability to input search specifications and have the matches displayed vertically.

3. Performance- Depending on the requirements of the API web performance will depend on the user.
4. Supportability- This will be a local system of a low population area so data matches for queries will be in the hundreds maybe less. The flexibility of the system will depend on the search functions made.

UI Requirements-

1. A search bar
2. A vertical scroller for results that don't fit on one page.
3. The search result is displayed with the photo, Name, Make, Model, Year, and Dealer.



Plan of Work-

So far it is going well I have been doing research into Web API and have been beginning the Java design of my program. So far, I have designed a starter program

with entities I manually put in myself along with a routine and simple select function to be able to view more details about the selected vehicle.

Sequence Diagrams

Sequence Diagram for searching for a car

Actor: A customer with a computer on the website

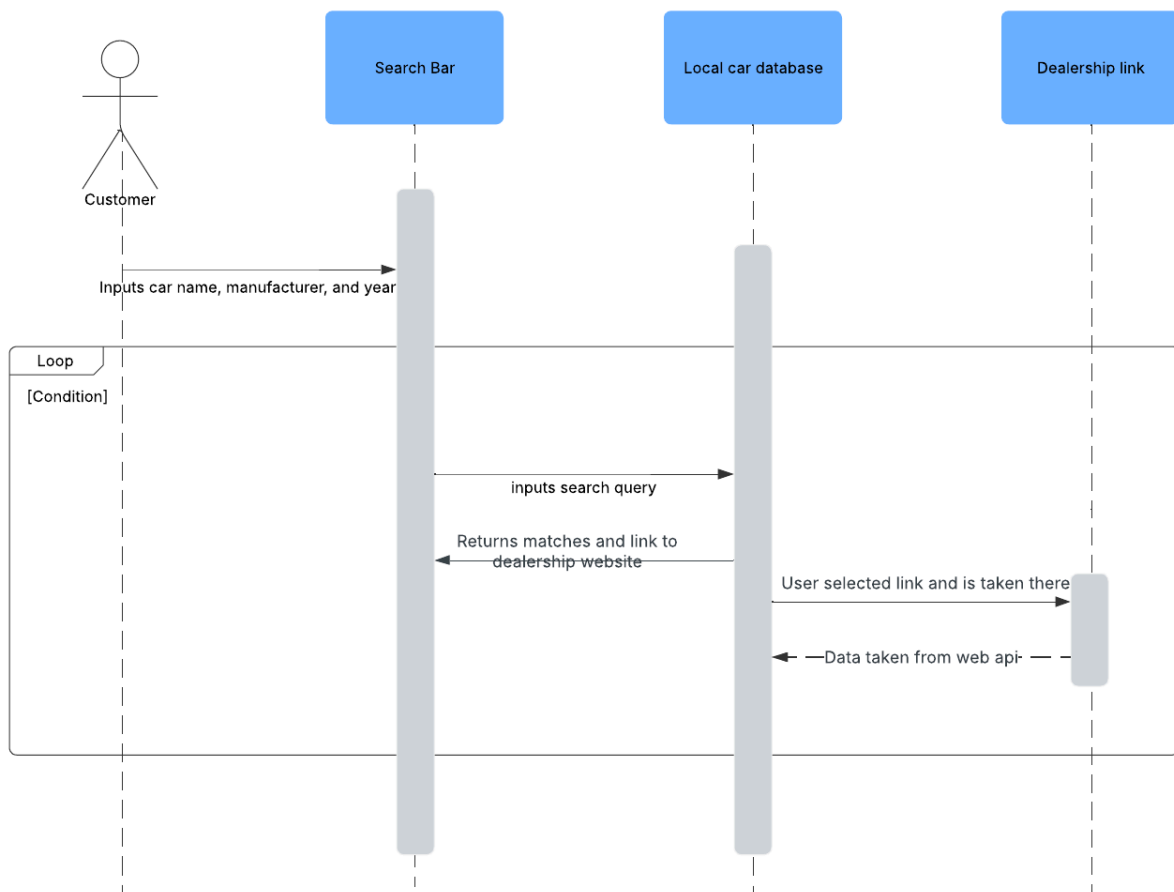
Object: Search Bar

Interacts with the Local Car Database and the Dealership web page (link)

Steps of searching for a car:

1. The customer navigates to my website and is greeted by the search bar
2. The customer inputs the make, model, and year (pending further filters)
3. The Search bar takes the query and uses the web API to search and match the customer's search.
4. The website displays the matching results vertically so that the customer can select and be directed to the web page of the car dealership.

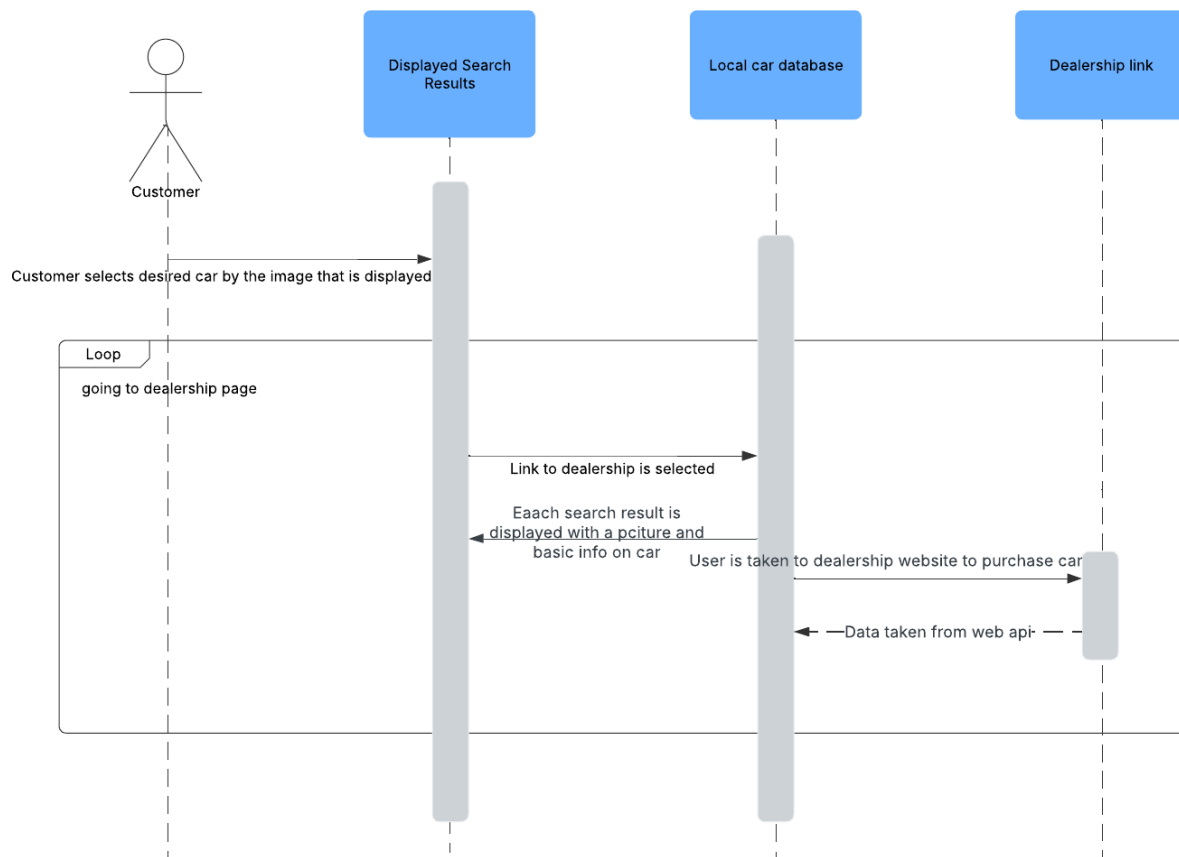
The sequence diagram is shown below:



Sequence Diagram for selecting a car:

1. The user searches for the desired car in the search bar
2. The user is shown matches displayed vertically with a picture and basic information on the vehicle
3. The user selects the desired match by clicking on the image of the desired car
4. The user is taken to the dealership webpage actually to purchase the vehicle or seek further information

The sequence is a diagram shown below:



Activity Diagrams

Activity Diagram for searching for a 2004 Ford Mustang (silver):

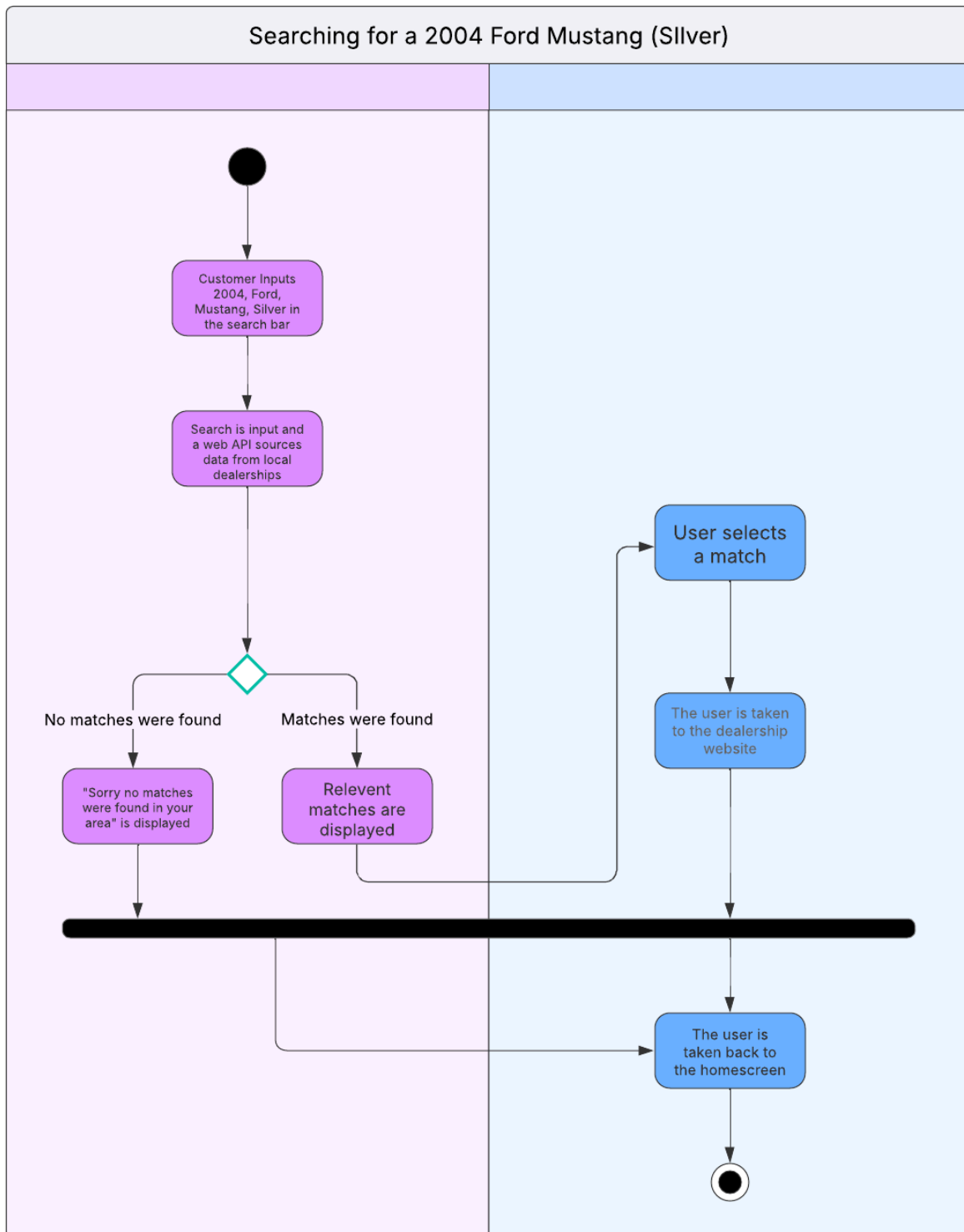
States

-Initial State: The customer navigates to my website

-Final State: 1. The customer finds the appropriate match and is taken to the dealership website. 2. There's no match in the local area and the user is prompted so and taken back to the homepage to enter another search.

Actions

The customer navigates to my website and is greeted by the homepage. The customer then inputs the search query "2004, Ford, Mustang, Silver" exactly and any matches from dealerships in the area will be displayed vertically with their relevant info and pictures. If the customer selects the picture of the desired vehicle they will be taken to the dealership page the data was sourced from. Based on the order above, the activity diagram for searching for a 2004 Ford Mustang Silver is given below:



Activity Diagram for navigating to match's dealership home web page to purchase the vehicle

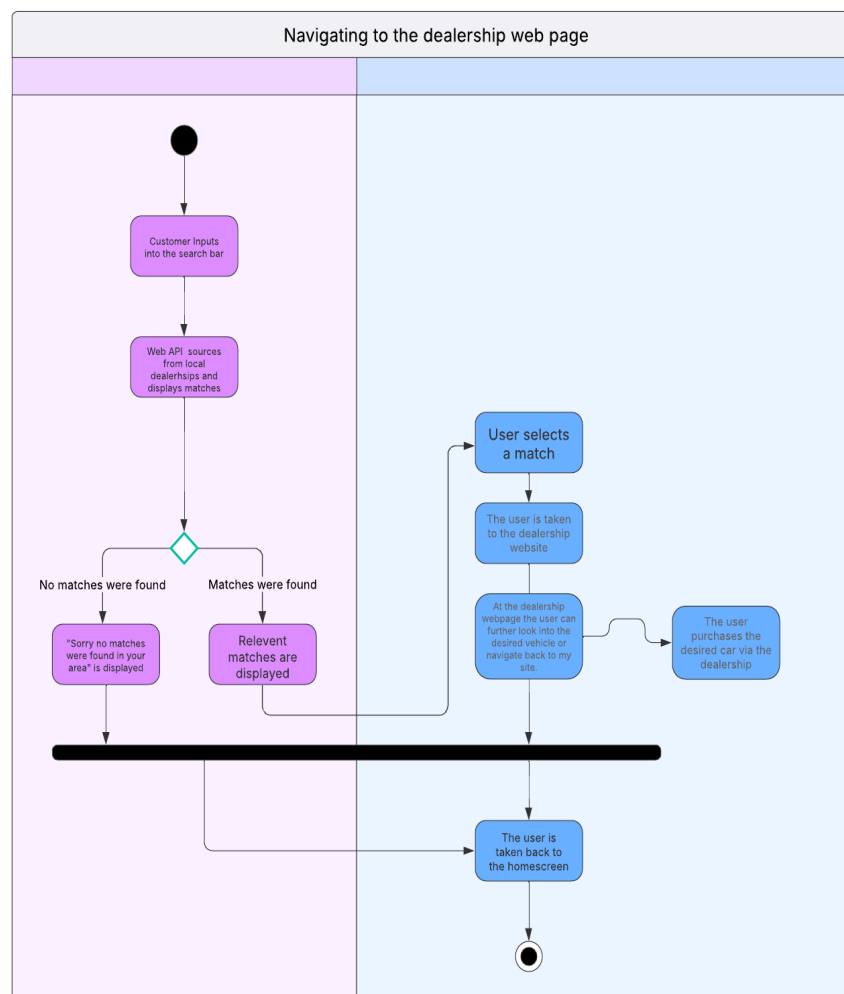
States

-Initial State: Customer searches for a desired vehicle in their area.

-Final State: The customer is brought to the dealership web page for the desired vehicle to purchase.

Actions

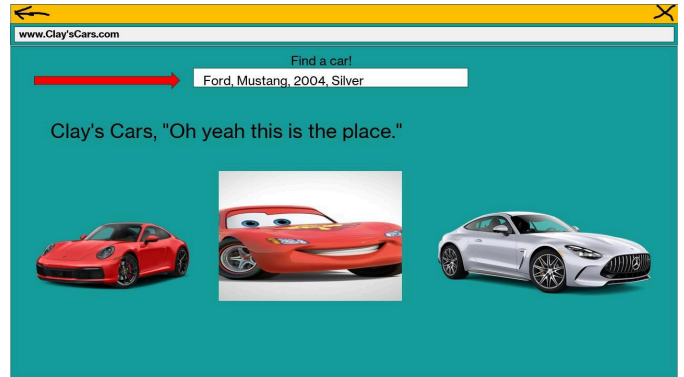
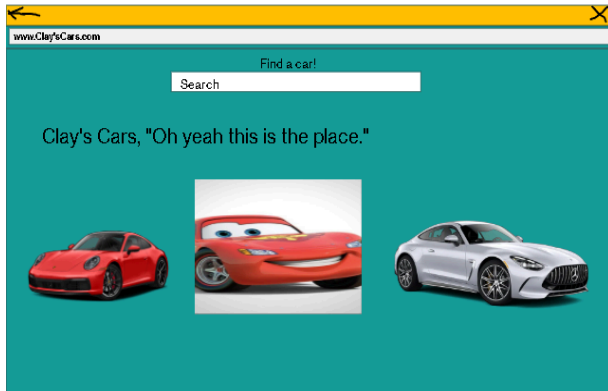
After the customer inputs a search query into the search bar on my websites, relevant matches will be displayed vertically with basic information such as make, model, year, and color as well as the pictures sourced from the dealership websites which will be used as the link icon to navigate to the dealership website. The activity diagram for this is shown below:



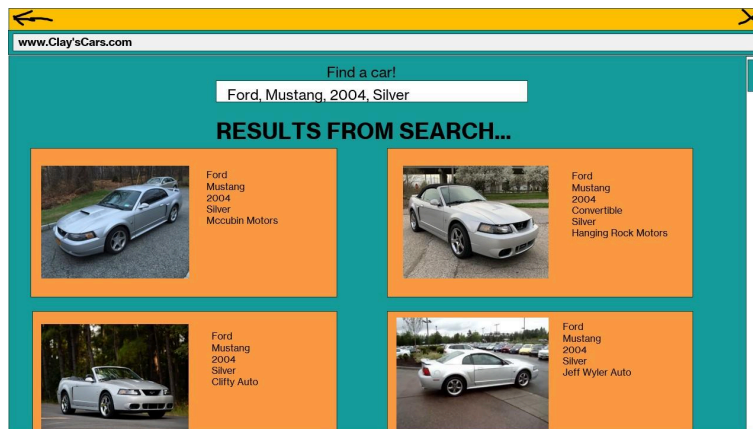
User Interface Specifications

Preliminary Design:

Use Case: A user searches for a Ford Mustang 2004, Silver in his area.

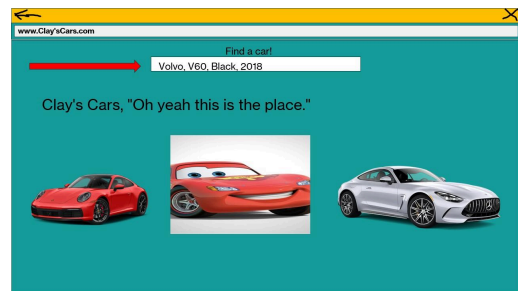
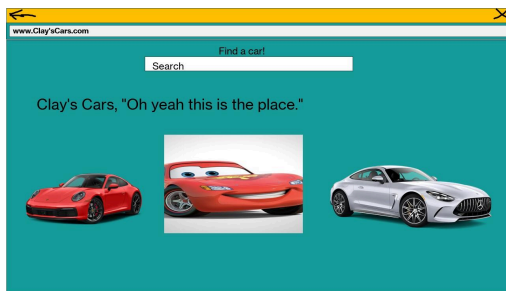


On the main page, the user clicks on the search barThe User inputs the search query

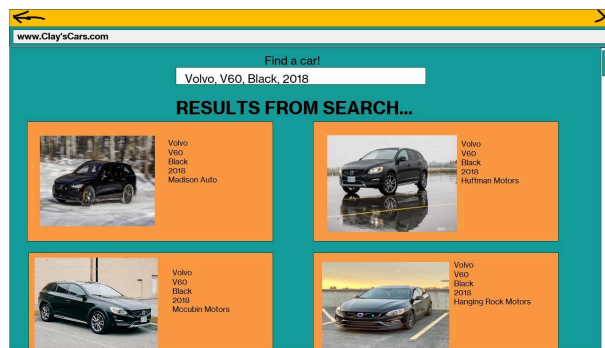


Results of the search query displayed

Use Case: A user searches for a Volvo, V60, Black, 2018

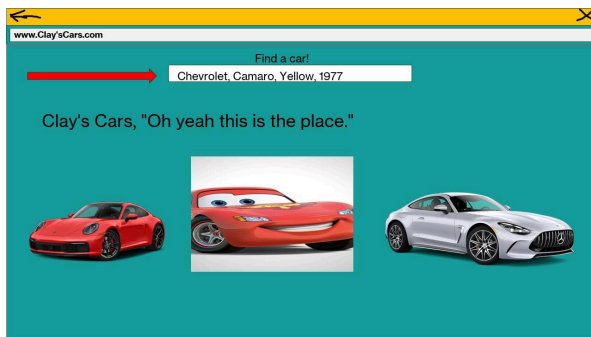


On the main page, the user clicks on the search bar
The User inputs the search query

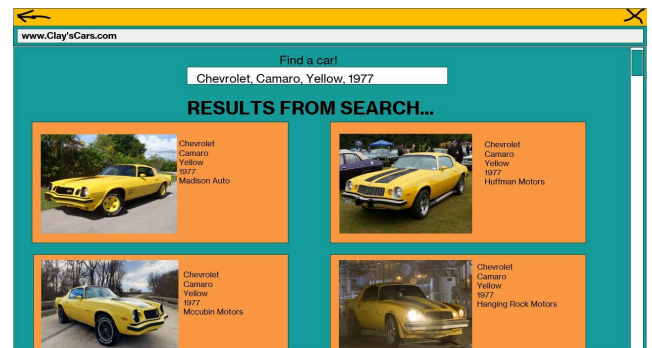


The results of the query are displayed

Use Case: A user searches for a Chevrolet, Camaro, Yellow, 1977

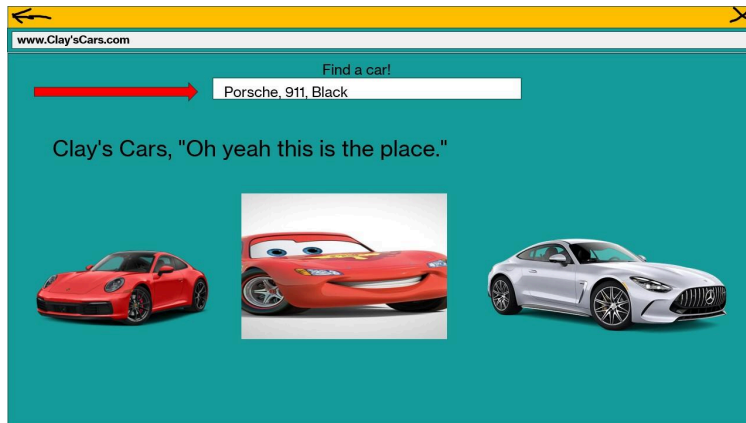


The user inputs the search query

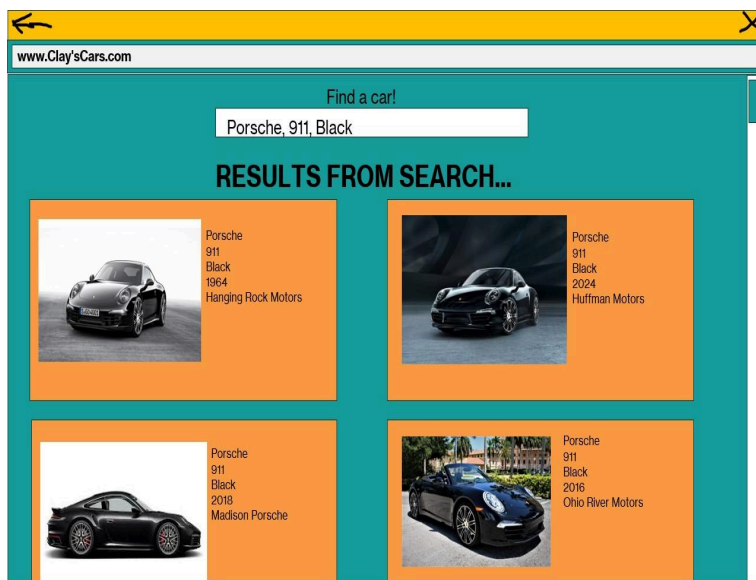


Result of the query is displayed

Use Case: A user searches for a Porsche, 911, Black

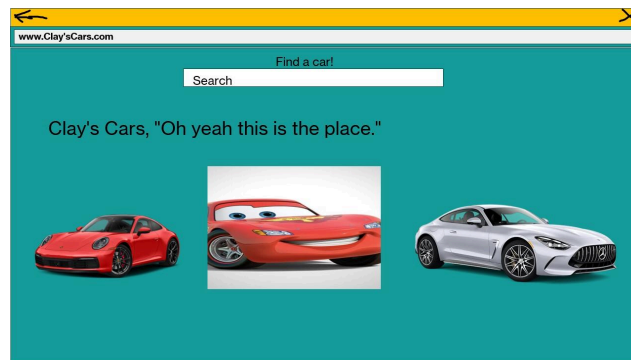


The user inputs the query in the search bar

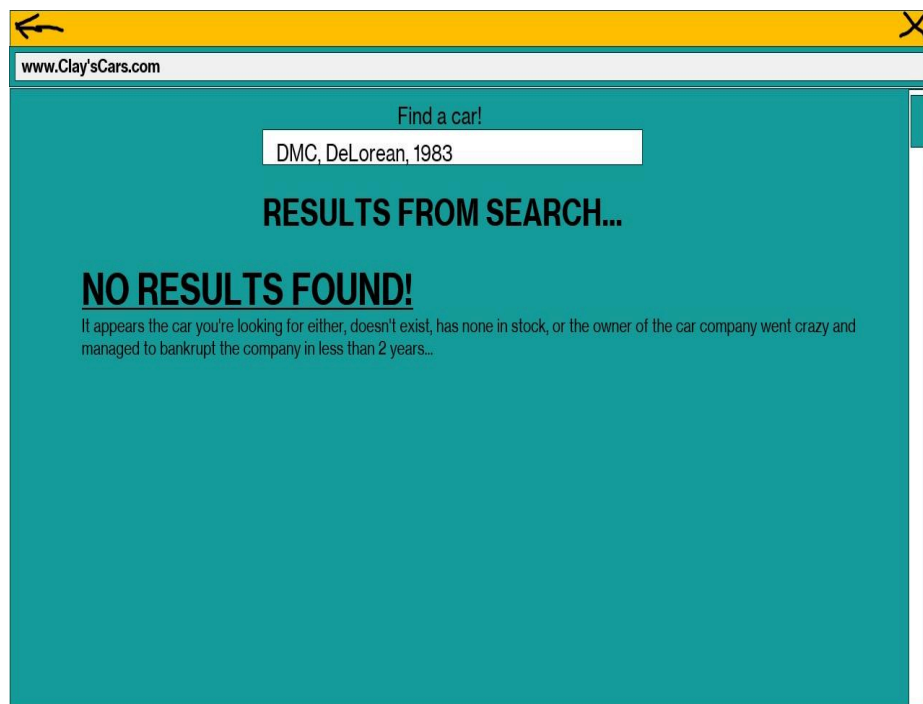


The results of the query are displayed

Use Case: A user searches for a car but no results are found



On the main page, the user clicks on the search bar to input a query



If no cars in my area fit your description, you will be greeted by this page and the only thing to do is go back to the main page or enter another query

Use Effort Estimation

Usage Scenario	Navigation	Keystrokes	Clicks
The user searches for a 2004 Silver Ford Mustang	Search bar, enter, select the desired result	<20	2
The user searches for a 2018 Black Volvo V60	Search bar, enter, select the desired result	<20	2
The user searches for a car but no results are found and the user returns to the main menu	Search bar, enter, back button	<20	3

Project Plan

- A fully accessible website via the internet
- A web API-integrated database of all cars available at my local dealerships
- A fully animated and decorated website
- An efficient search engine and search display as conceptualized
- A function in the search results to take the user to the home dealership of the car selected
- Motorcycle and Facebook Marketplace integration?